

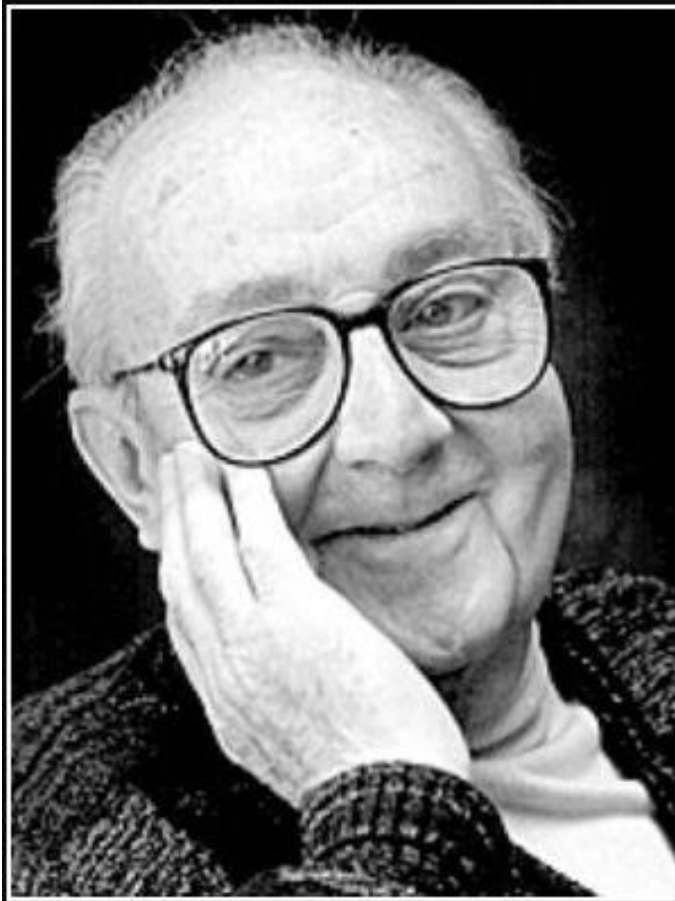
Model Checking

Announcements

- Chapter 6 and 7 today

- Final Project
 - Presentations

All models are wrong

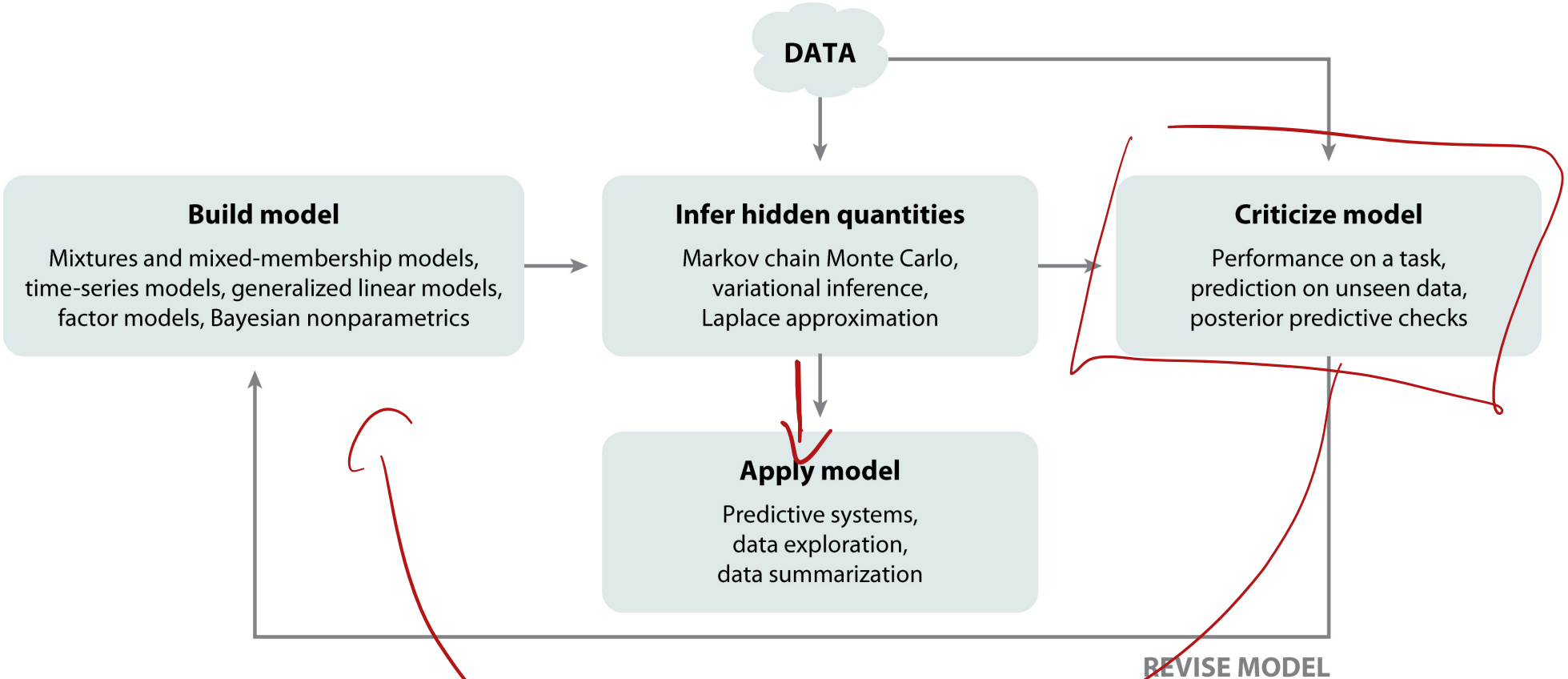


All models are wrong, but some are
useful.

— *George E. P. Box* —

If the model is “wrong”, how can we improve it?

Box's Loop



AR Blei DM. 2014.
Annu. Rev. Stat. Appl. 1:203–32

Posterior predictive model checking

(PPC)

- Let y_{obs} represent the observe data $y_1, \dots, y_n$
- Let $\tilde{y}$ represent n replicated (e.g fake) observations generated from the model

Mimic
original
data.

- $p(\tilde{y} \mid y_{\text{obs}}) = \int p(\tilde{y} \mid \theta) p(\theta \mid y_{\text{obs}}) d\theta$

- Generate test quantity from $t(\tilde{y})$
- Check if the simulated test quantities are similar to the observed test quantity, $t(y_{\text{obs}})$

variance
quantiles etc.

$t(\tilde{y})$

$\gamma / \lambda \sim \text{Pois}(\lambda)$
 $p(\lambda) \sim \text{Gam}$
 $\Rightarrow p(\lambda / \gamma) \sim \text{Gam.}$

$\Rightarrow \text{P.P. } p(\tilde{\gamma} / \gamma) \sim \int p(\tilde{\gamma} / \lambda) p(\lambda / \gamma) d\lambda$

$\sim \text{Negative-Binomial.}$

$\gamma \sim \text{Bin}$

$\Theta \sim \text{Beta}$

$\tilde{\gamma} / \gamma \sim \text{Beta-Binom.}$

Generate $\tilde{\gamma}$ via Monte-Carlo.

- MC proc.
1. Get a sample $\Theta^s \sim p(\Theta / y_{\text{obs}})$
 2. Generate $\tilde{\gamma}^s / \Theta^s \sim p(\tilde{\gamma} / \Theta^s)$



Posterior predictive model checking

- If the model fits the data, then fake data generated under the model should look similar to the observed data
- Discrepancies can be due to model misfit or chance (or both!)
- Monte Carlo approach:
 1. sample $\theta^{(s)} \sim p(\theta | Y = y_{\text{obs}})$
 2. sample $\tilde{y}^{(s)} = (\tilde{y}_1^{(s)}, \dots, \tilde{y}_n^{(s)}) \sim \text{i.i.d. } p(y | \theta^{(s)})$
 - $\tilde{y}$ has same number of observations as y_{obs}
 3. compute $t^{(s)} = t(\tilde{y}^{(s)})$

A Bayesian Modeling Process (overview)

1. Propose a sampling model and prior distributions

2. Compute the posterior distribution, here

$$p(\theta \mid Y = y) \sim \text{Gamma}(a + y, \beta + \nu)$$

3. Simulate test statistics, $T(\tilde{y})^{(s)}$ from the posterior predictive distribution

- for s in $1:\text{num_fake_data}$

- Sample $\theta^{(s)} \sim \text{Gamma}(a + y, b + \nu)$

- Sample $\tilde{y}^{(s)} \sim \text{i.i.d Pois}(\theta^{(s)})$ (same sample size as y_{obs})

- Compute $T(\tilde{y}^{(s)})$

$y \sim \text{Pois}(\lambda)$

$\lambda \sim \text{Gam.}$

4. Compare the samples $T(\tilde{y}^{(s)})$ to $T(y_{\text{obs}})$
5. Identify any model misfit, go back to step 1 and repeat.

Choosing Test Statistics

$$P(T(\tilde{y}, \theta) | y_{obs})$$

- Test quantity, or discrepancy measure, $T(\tilde{y}, \theta)$ can also be a function of parameters
- A model can be inadequate for some purposes but adequate for others
- Avoid choosing sufficient statistics
- Statistics should be relevant to the scientific purpose
- Often measures a feature of the data not directly addressed by the probability model
 - e.g. Ranks of the sample or correlation of residuals

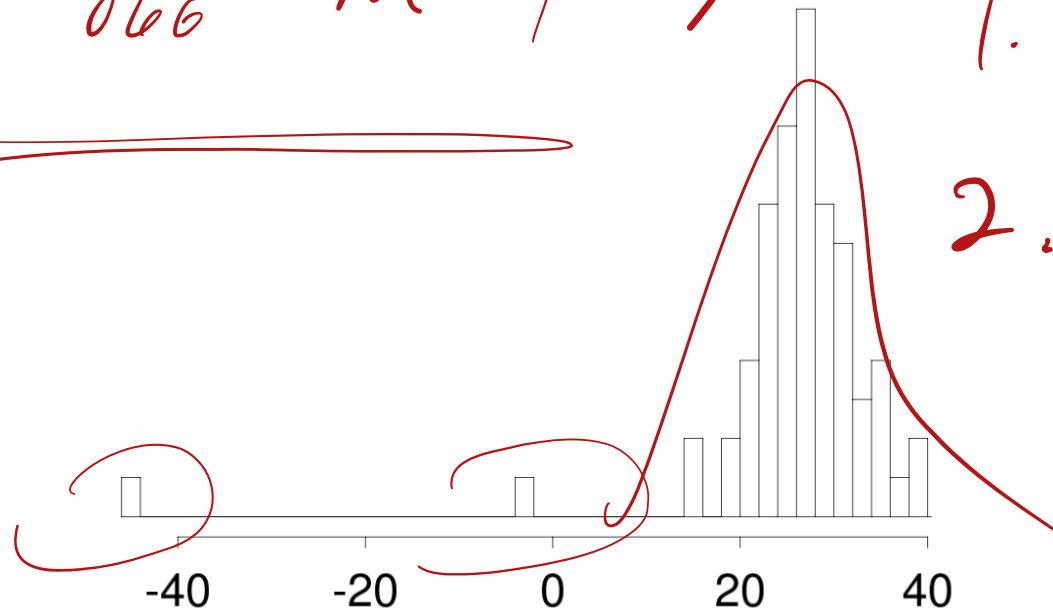
$$y_i \sim \text{Pois}(\lambda)$$

$$\bar{y}^{obs} \text{ vs } \bar{y}^{ref.}$$

Speed of Light Example

- Newcomb measured the amount of time required for light to travel a distance of 7442 meters.
- 66 measurements were collected

$$y_1, \dots, y_{66} \sim \mathcal{N}(\mu, \sigma^2)$$



$$P(\mu, \sigma^2 | y_1, \dots, y_{66})$$

1. $\mu^s, \sigma^2 \sim \text{posterior}$

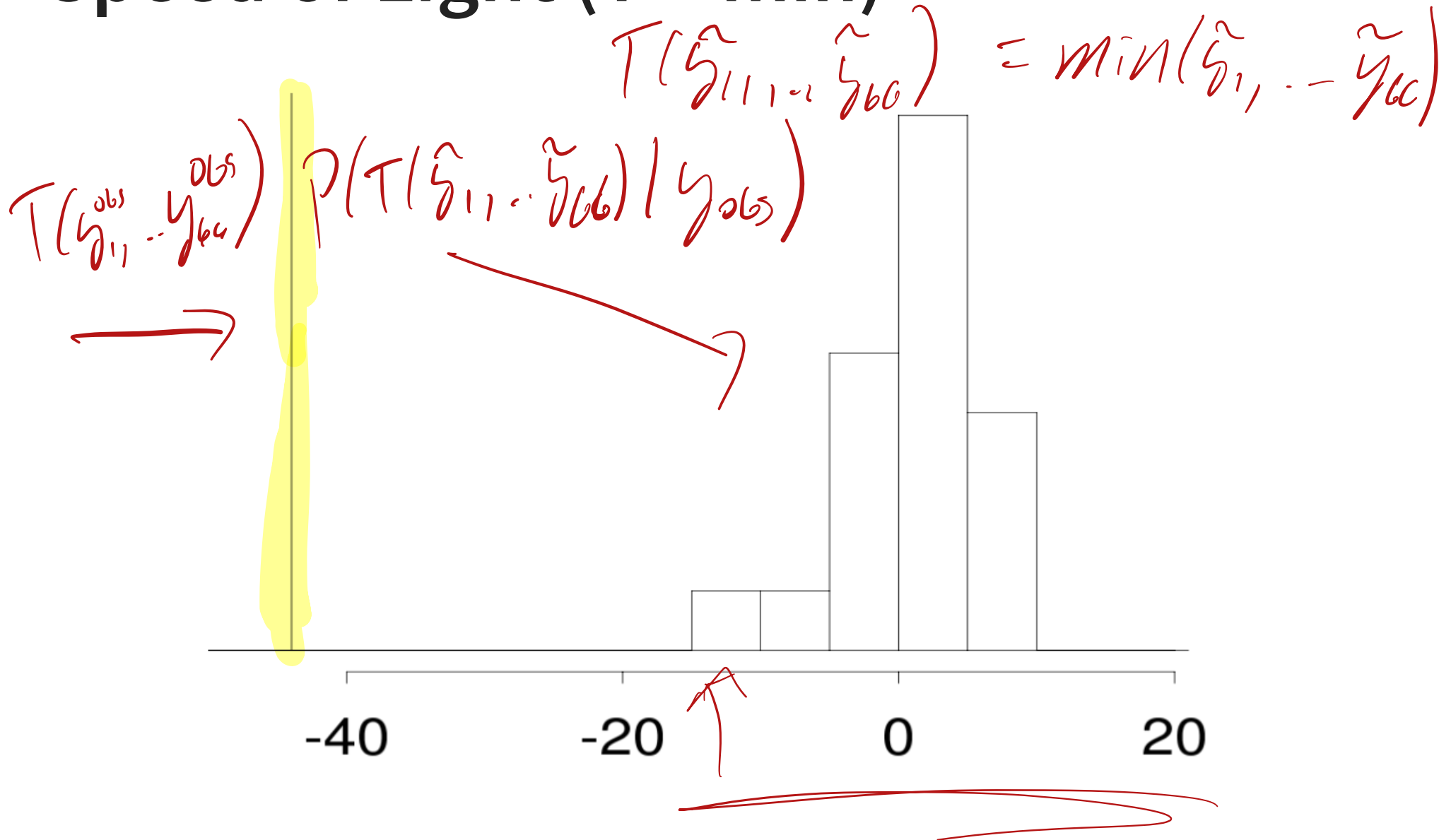
2. Generate

$$\tilde{y}_1, \dots, \tilde{y}_{66} \mid$$

$$\mu^s, \sigma^{2s}$$

3. $T(\tilde{y}_1, \dots, \tilde{y}_{66})$

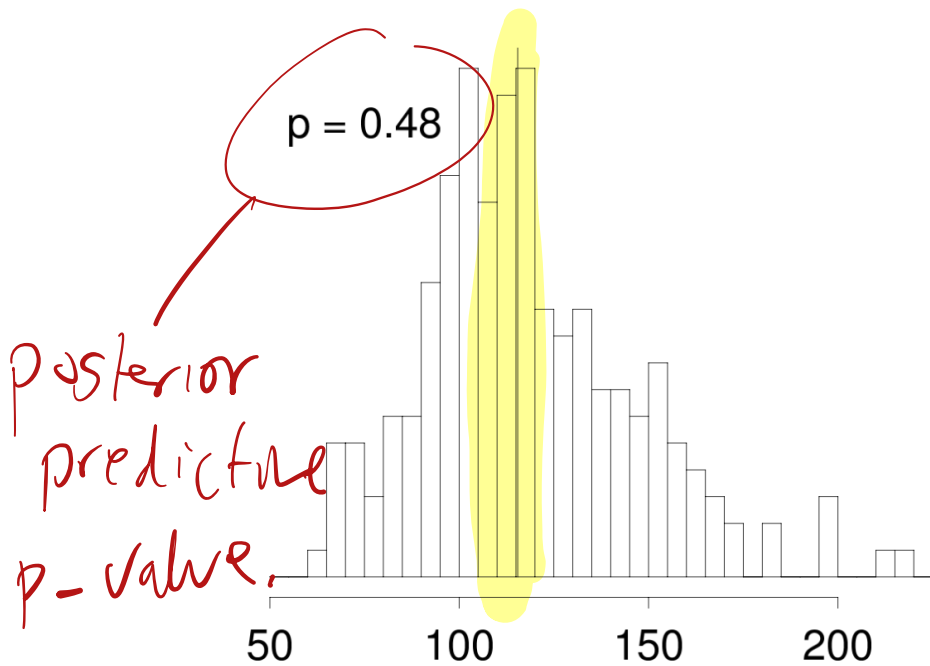
Speed of Light ($T = \min$)



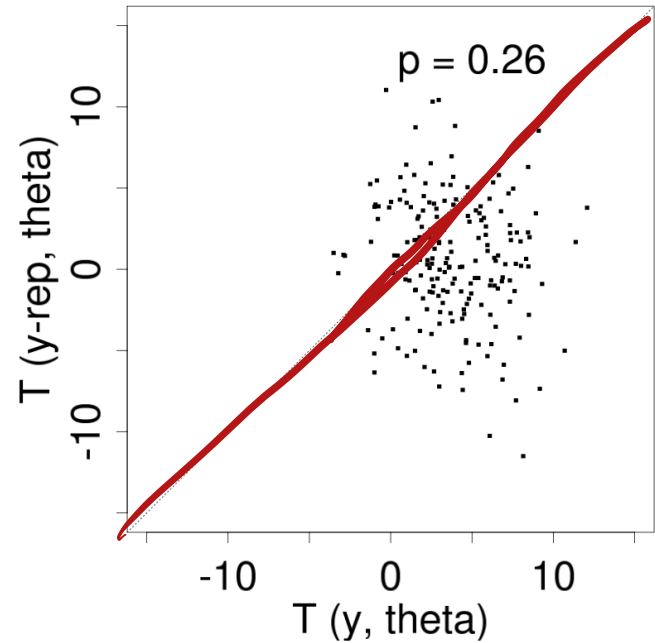
Speed of Light Example

$$P(\text{var}(\hat{\gamma}_1, \dots, \hat{\gamma}_{66}) | y_{\text{obs}})$$

$$\begin{aligned} & \frac{|\hat{\gamma}_{(61)} - \mu|}{|\hat{\gamma}_{(6)} - \mu|} \xrightarrow{90\%} \frac{10\%}{\text{quantile}} \end{aligned}$$



$$T(\hat{\gamma}, \mu)$$



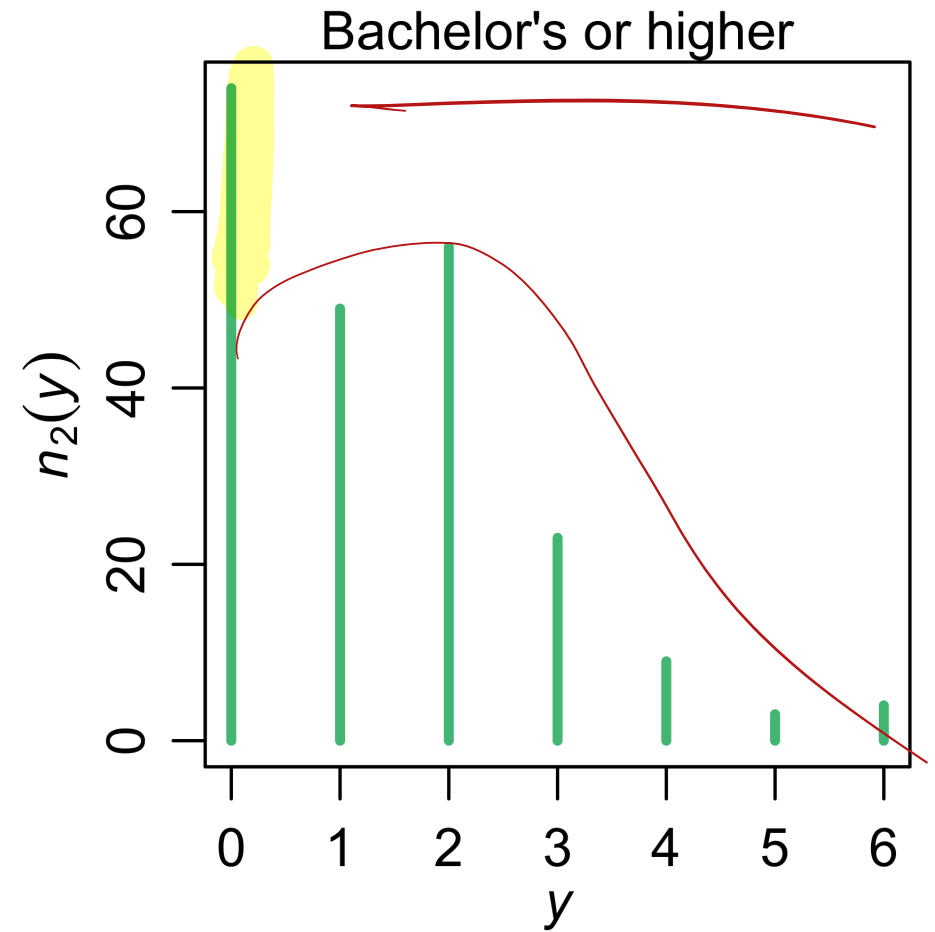
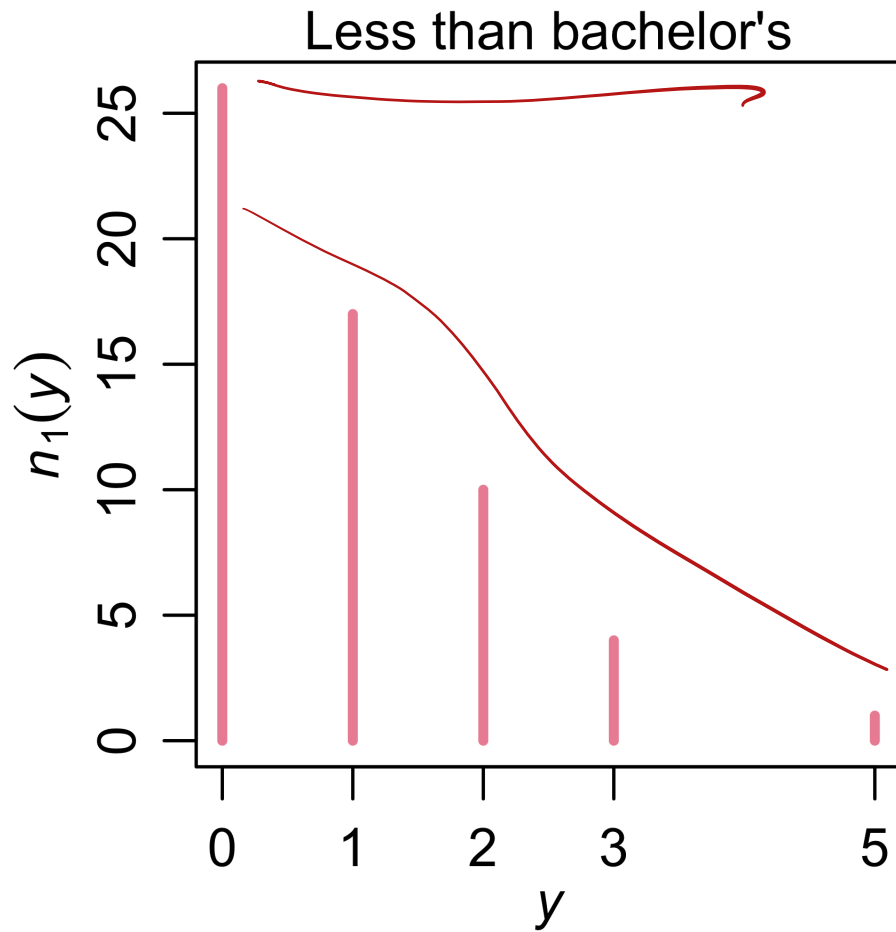
$$T(y^{\text{obs}}, \mu)$$

Predictive Checks: an example

- In the 1990's there was a survey of 155 women, at least 40 years of age
- Recorded number of children and educational attainment
 - Bachelor's degree or higher ($n_1 = 111$)
 - Less than bachelor's degree ($n_2 = 44$)

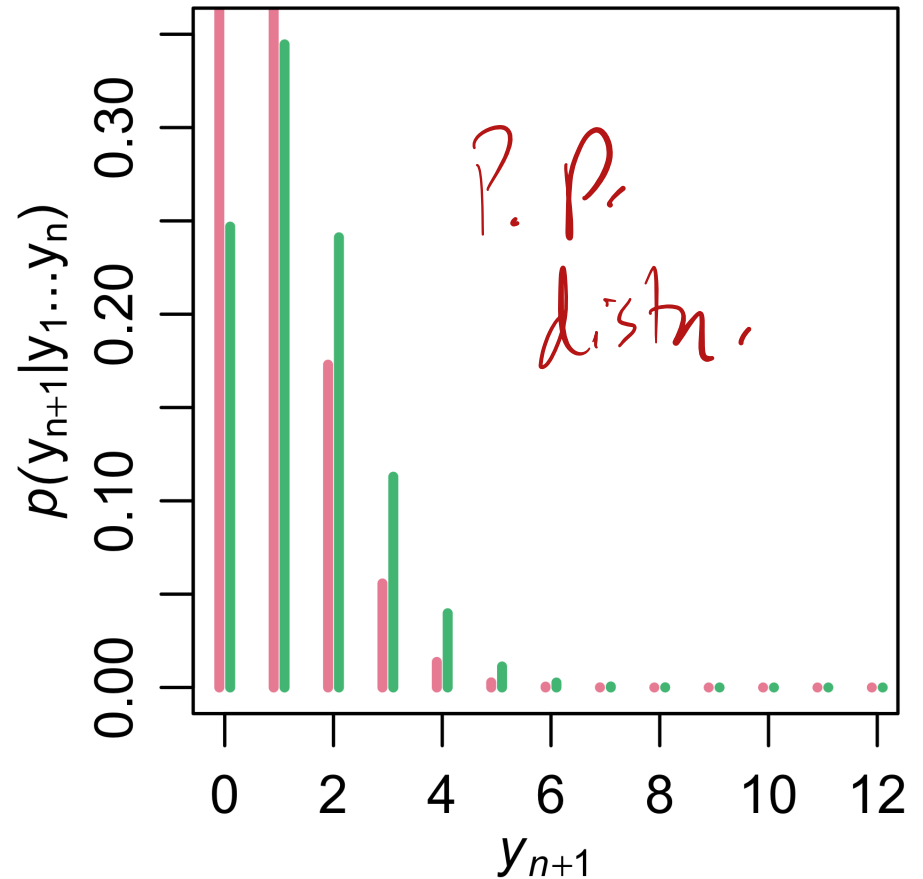
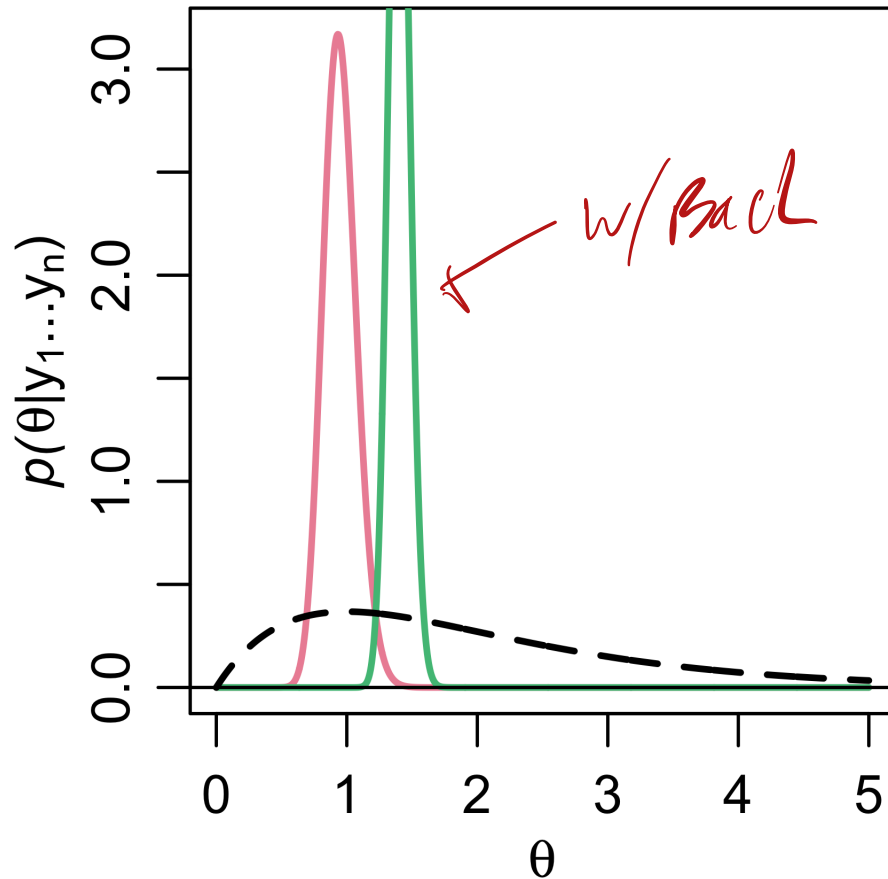
$$Y_{1,1} \dots, Y_{n_1,1} | \theta_1 \sim \text{i.i.d. Poisson}(\theta_1) \quad \text{Bach.}$$
$$Y_{1,2} \dots, Y_{n_2,2} | \theta_2 \sim \text{i.i.d. Poisson}(\theta_2) \quad \text{No Bach}$$

PPCs example



PPCs example

$$P(X | \dots)$$



PPCs example

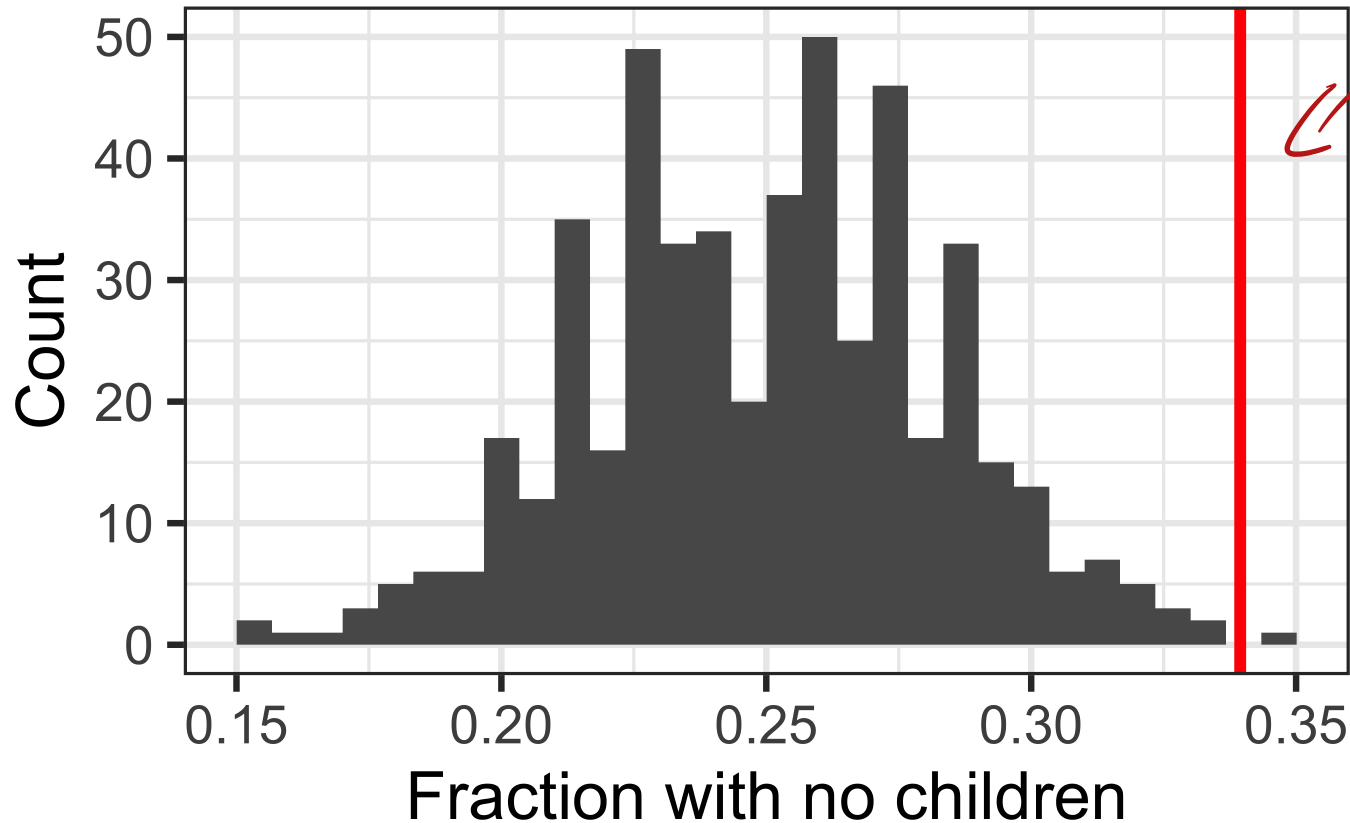
- Let's check the model fit for the “without Bachelor's” group first
- Do S times:
 - sample $n_2 = 44$ observations $\tilde{y}$ from the posterior predictive distribution
- Let $T(\tilde{y})$ be the fraction of women with no children

```
1 S <- 1000
2 t_s <- numeric(S)
3 for(s in 1:S){
4   theta_s <- rgamma(1,  $\tilde{a}$ ,  $\tilde{b}$ ) # whatever a and b are for my posterior
5   ytilde_s <- rpois(n=44, theta = theta_s)
6   t_s[s] <- mean(ytilde_s == 0) # compute test stat
7 }
8
9 ## then visualize histogram of t_s
```

$\lambda \sim \text{Com}(\tilde{y} + a, n + b)$

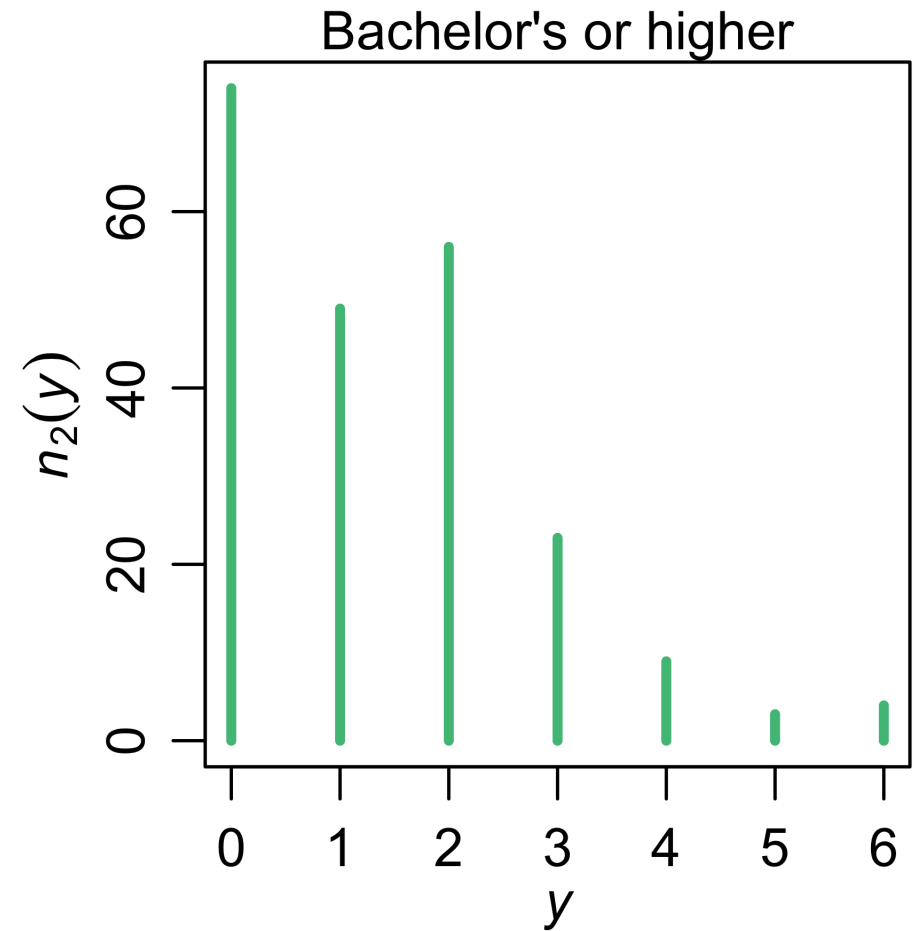
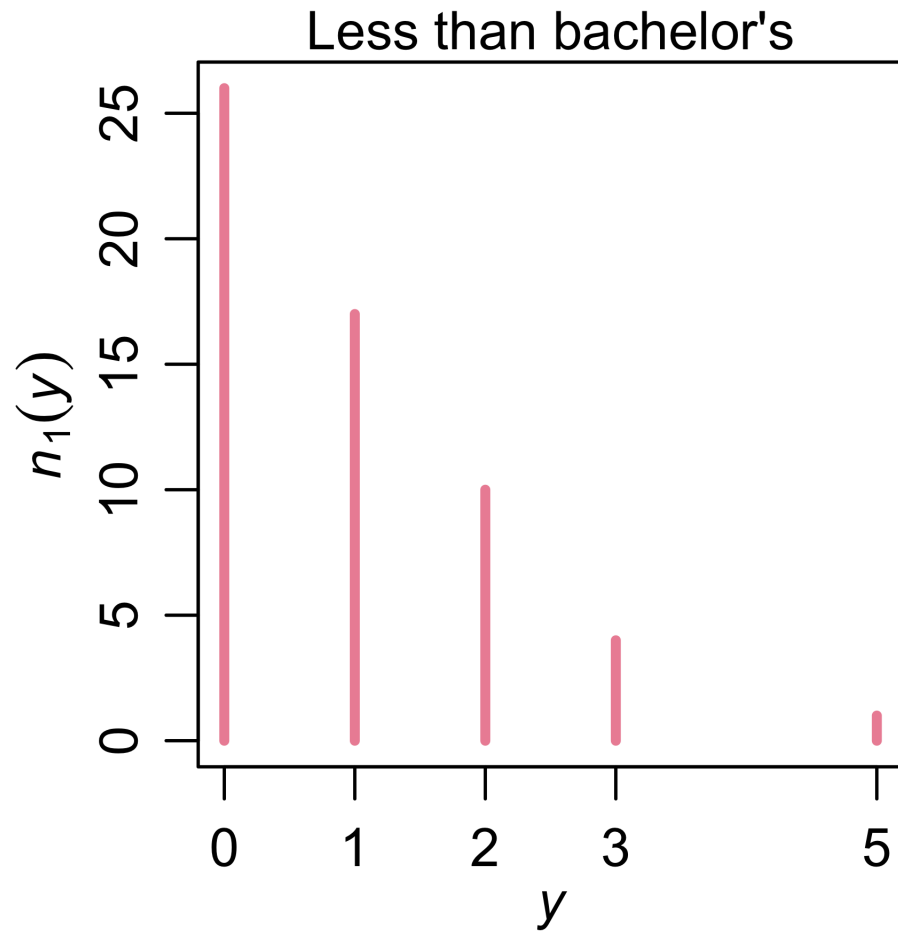
PPCs example

Women without Bachelors degree



$$Pr((T^{rep} > T^{obs} \mid y_{obs}) = 0.002$$

PPCs example



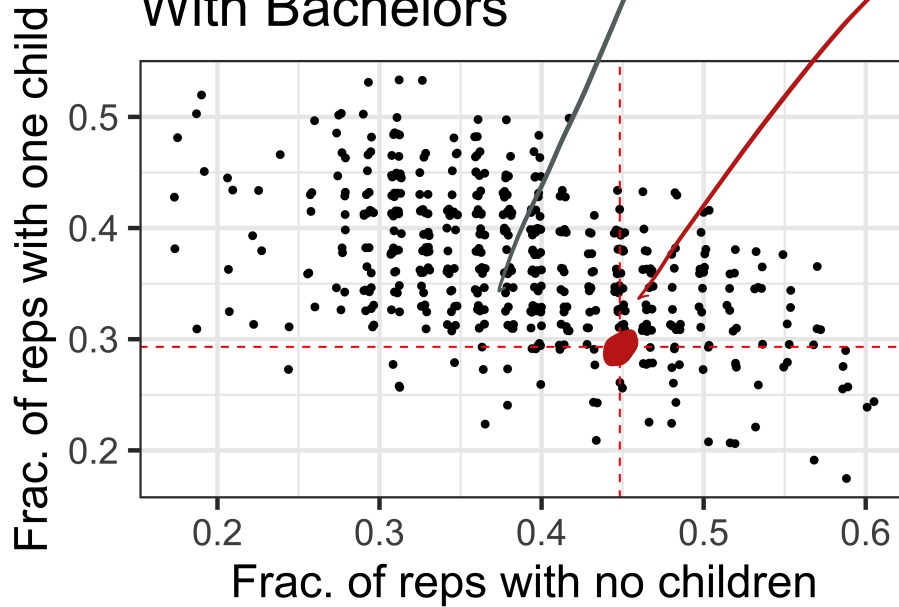
PPCs example

- Model checking both groups
- Look at fit for two different test statistics:
 - Fraction with no children ✓
 - Fraction with one child ✓

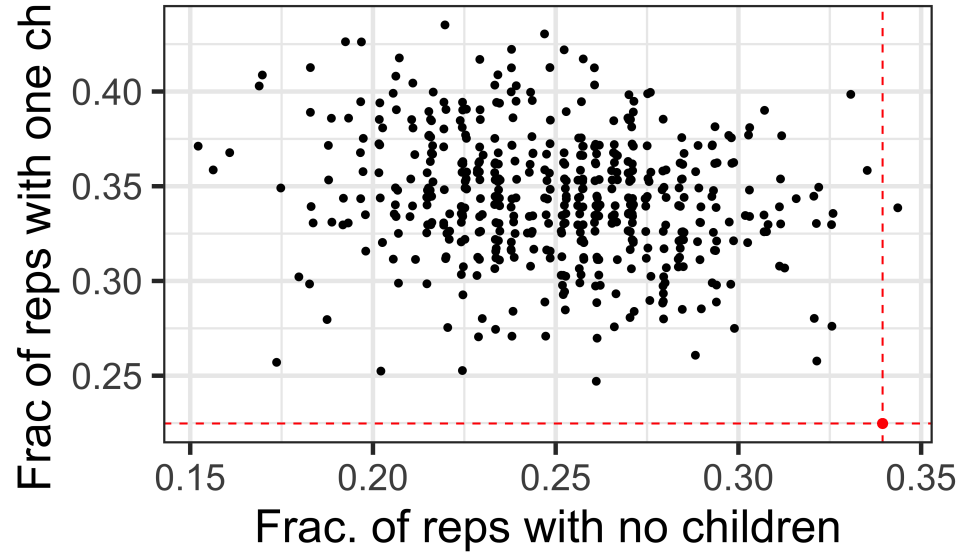
Poisson example

$$P(\tau_1(y_{obs}), \tau_2(y_{obs}) \mid y^{obs})$$

With Bachelors



τ_{obs} Without Bachelors



Predictive Distributions in Stan

```
1  ...
2  } model {
3    y1 ~ poisson(lambda1);
4    y2 ~ poisson(lambda2);
5  } generated quantities{
6
7    int y1rep[n1];
8    int y2rep[n2];
9    real log_lik[n1+n2];
10   for(i in 1:n1){
11     y1rep[i] = poisson_rng(lambda1);
12     log_lik[i] = poisson_lpmf(y1[i] | lambda1);
13   }
14   for(i in 1:n2){
15     y2rep[i] = poisson_rng(lambda2);
16     log_lik[n1+i] = poisson_lpmf(y2[i] | lambda2);
17   }
18 }
```

data

y

Predictive Checks in Stan

Running MCMC with 4 sequential chains...

Chain 1 finished in 2.0 seconds.

Chain 2 finished in 2.1 seconds.

Chain 3 finished in 2.3 seconds.

Chain 4 finished in 2.0 seconds.

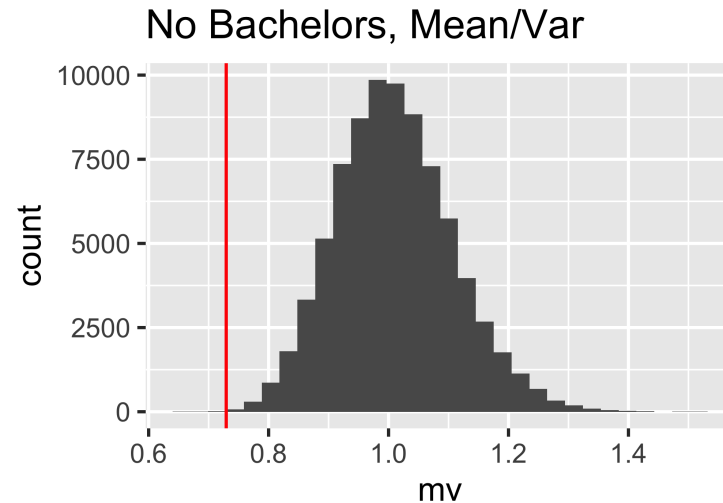
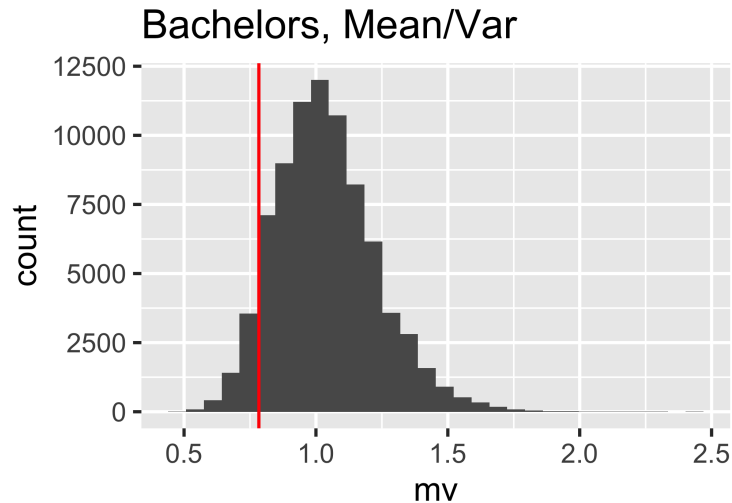
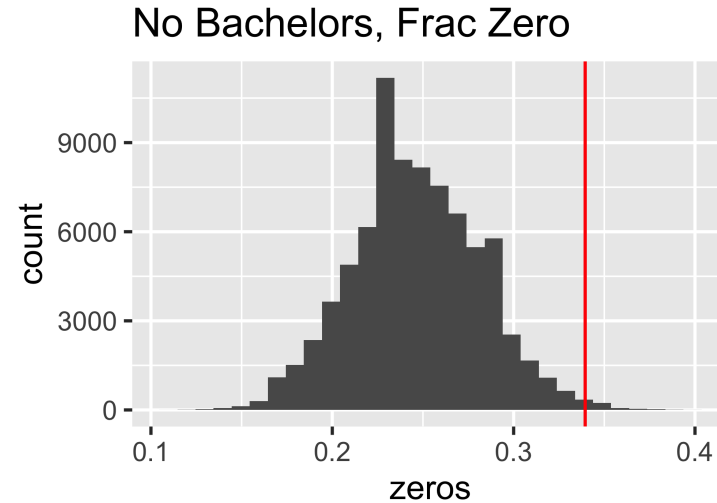
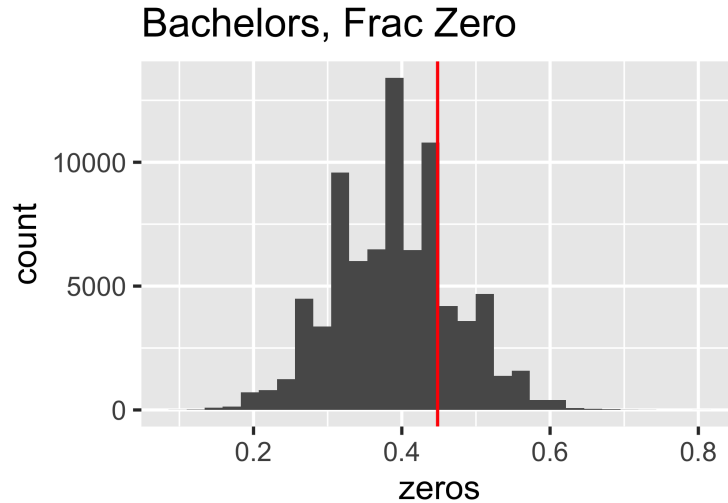
All 4 chains finished successfully.

Mean chain execution time: 2.1 seconds.

Total execution time: 8.5 seconds.

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp__	-261.16	-260.85	1.02	0.71	-263.21	-260.21	1.00	35823	47636
lambda1	0.95	0.94	0.13	0.13	0.75	1.17	1.00	75976	54495
lambda2	1.40	1.40	0.08	0.08	1.27	1.54	1.00	75256	53473
y1rep[1]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	80952	80389
y1rep[2]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	79270	79427
y1rep[3]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	79299	78493
y1rep[4]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	78656	80525
y1rep[5]	0.95	1.00	0.98	1.48	0.00	3.00	1.00	78714	78498
y1rep[6]	0.96	1.00	0.99	1.48	0.00	3.00	1.00	80345	79532
y1rep[7]	0.94	1.00	0.98	1.48	0.00	3.00	1.00	79840	79077
:	:	:	:	:	:	:	:	:	:

Predictive Checks (Pois. Model)



ZIP

Running MCMC with 4 sequential chains...

Chain 1 finished in 0.4 seconds.

Chain 2 finished in 0.4 seconds.

Chain 3 finished in 0.5 seconds.

Chain 4 finished in 0.5 seconds.

All 4 chains finished successfully.

Mean chain execution time: 0.5 seconds.

Total execution time: 2.0 seconds.

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp__	-423.61	-423.24	1.54	1.28	-426.63	-421.84	1.00	1454	2020
lambda1	1.20	1.18	0.21	0.21	0.89	1.58	1.00	2372	2583
lambda2	1.75	1.75	0.13	0.13	1.54	1.96	1.00	2604	2453
theta1	0.21	0.21	0.10	0.11	0.04	0.38	1.00	2053	1306
theta2	0.20	0.20	0.04	0.04	0.13	0.27	1.00	2503	1935
ylrep[1]	0.94	1.00	1.12	1.48	0.00	3.00	1.00	3939	3904
ylrep[2]	0.96	1.00	1.11	1.48	0.00	3.00	1.00	3790	3780
ylrep[3]	0.94	1.00	1.10	1.48	0.00	3.00	1.00	3330	3069
ylrep[4]	0.92	1.00	1.09	1.48	0.00	3.00	1.00	3215	3372
ylrep[5]	0.94	1.00	1.10	1.48	0.00	3.00	1.00	3984	3862

ZIP

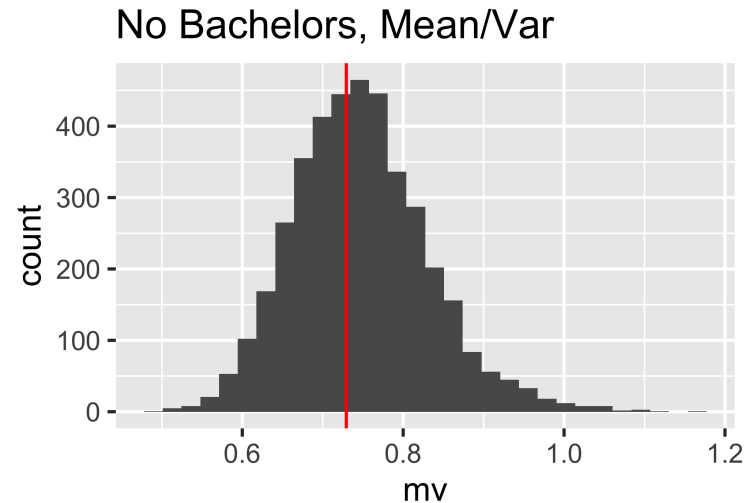
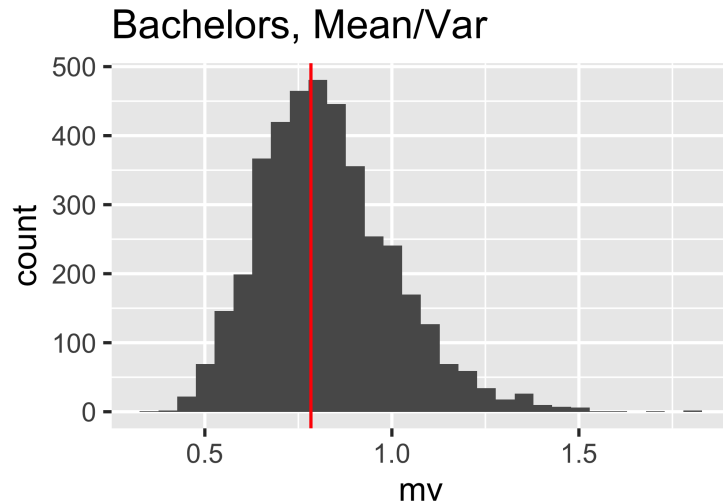
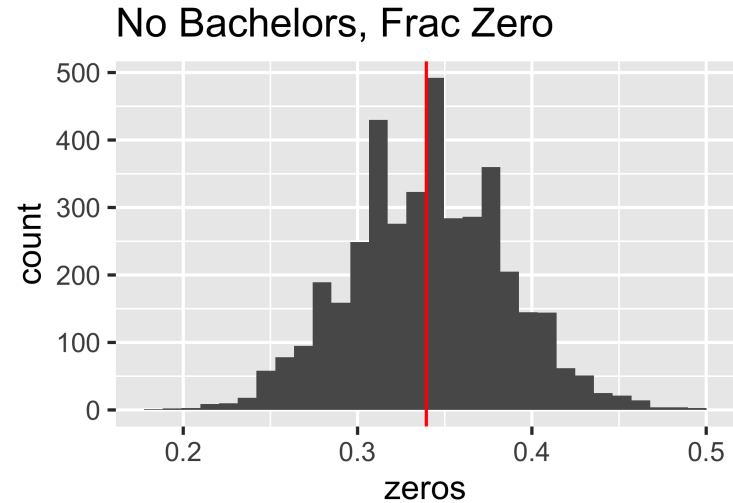
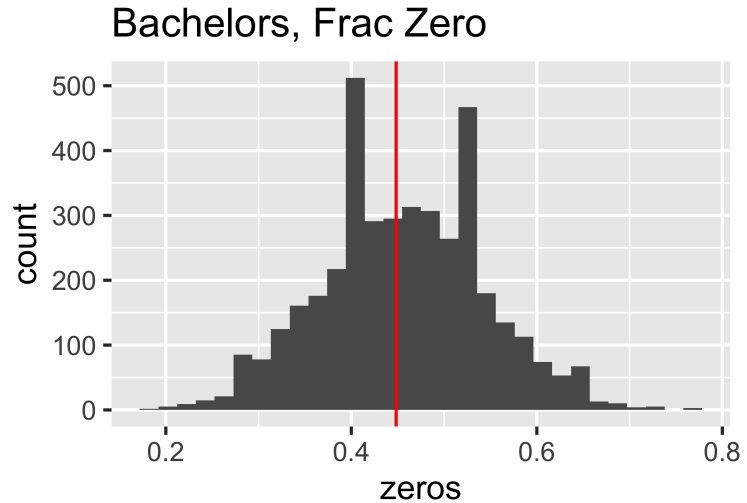
$$Z \sim \text{Bern}(\phi)$$

$$Y = 0 \quad \text{if } Z = 1$$

$$Y \sim \text{Pois}(\lambda) \quad \text{if } Z = 0$$

$$P(Y|\lambda) = \phi I[Y=0] + (1-\phi) \frac{e^{-\lambda} \lambda^Y}{Y!}$$

Predictive Checks (ZIP Model)



Model Comparison

- PPCs for checking model fit. But how do we compare different models (regardless of fit)?
- Compare models by their out-of-sample predictive power
- Leave-one-out cross validation
 - Fit model n times, leaving out one observation
 - Evaluate predictive accuracy on left out observations
 - Typically used for parameter tuning
- Can use ideas from LOO-CV to compare models

Measures of Predictive Accuracy

- Point prediction: predict a single unknown future observation
- Measures of predictive accuracy are called *scoring functions* (e.g. MSE)
- Probabilistic predictions account for uncertainty
 - **Scoring rules:** measure of accuracy based on probabilistic predictions
 - Examples: quadratic, logarithmic and zero-one scores

Log-likelihood as a scoring rule

- As a measure of accuracy, we call it the “log predictive density”
- $\log(p(y|\theta))$ is proportional to MSE if the data are normal with constant variance
- Connection to Kullback-Leibler divergence
 - KL: $-E_p \log(q(y)/p(y)) = -E_p \log(q(y)) + E_p \log(p(y))$
 - Asymptotically, the lowest KL model is the one with the highest expected log predictive density

Out-of-sample prediction

Expected Log Pointwise Predictive Density

Pareto-Smoothed Importance Sampling

- When the data are conditionally independent,

$r_i^s = \frac{1}{p(y_i|\theta^s)} \propto \frac{p(\theta^s|y_{-i})}{p(\theta^s|y)}$ to get importance sampling estimates

- $p(\tilde{y}_i | y_{-i}) \approx \frac{\sum_{s=1}^S r_i^s p(\tilde{y}_i | \theta^s)}{\sum_{s=1}^S r_i^s}$
- $p(y_i | y_{-i}) \approx \frac{1}{\frac{1}{S} \sum_{s=1}^S \frac{1}{p(y_i|\theta^s)}}$

<https://link.springer.com/article/10.1007/s11222-016-9696-4>

```

1 library(loo)
2 nb_model <- cmdstan_model("nb_model.stan")
3 nb_fit = nb_model$sample(
4     data=list(n1=num_bachelors, n2=num_no_bachelors,
5     y1=bachelors_data, y2=no_bachelors_data),
6     refresh=0)

```

Running MCMC with 4 sequential chains...

Chain 1 finished in 0.4 seconds.
 Chain 2 finished in 0.4 seconds.
 Chain 3 finished in 0.4 seconds.
 Chain 4 finished in 0.5 seconds.

All 4 chains finished successfully.
 Mean chain execution time: 0.4 seconds.
 Total execution time: 1.8 seconds.

```

1 loo_compare(list("pois"=pois_fit$loo(),
2     "zip"=zip_fit$loo(),
3     "nb"=nb_fit$loo()))

```

	elpd_diff	se_diff
zip	0.0	0.0
nb	-3.2	2.4
pois	-8.6	4.6

Galaxies Example

- Velocities in km/sec of 82 galaxies (Corona Borealis region).
- Multimodality in such surveys is evidence for voids and superclusters in the far universe.
- Statistical question: how many clusters are there in this dataset?

Model Comparison

```
1 library("loo")
2 galaxy_speeds <- log(MASS::galaxies)
3 galaxy_results <- list()
4
5 ## Run model for each of k clusters
6 mix_model <- cmdstanr::cmdstan_model("mix_model.stan")
7 for(k in 1:5) {
8   galaxy_results[[k]] <- mix_model$sample(data=list(K=k, N=length(galaxy_sp
9                                           refresh=0, show_messages=FALSE)
10 }
```

Warning: 63 of 4000 (2.0%) transitions ended with a divergence.

See <https://mc-stan.org/misc/warnings> for details.

Warning: 16 of 4000 (0.0%) transitions hit the maximum treedepth limit of 10.

See <https://mc-stan.org/misc/warnings> for details.

Model Comparison

```
1 loo_compare(galaxy_results[[1]]$loo(),  
2             galaxy_results[[2]]$loo(),  
3             galaxy_results[[3]]$loo(),  
4             galaxy_results[[4]]$loo(),  
5             galaxy_results[[5]]$loo())
```

	elpd_diff	se_diff
model3	0.0	0.0
model5	-0.4	1.4
model4	-1.0	0.7
model2	-10.7	3.7
model1	-41.1	8.7